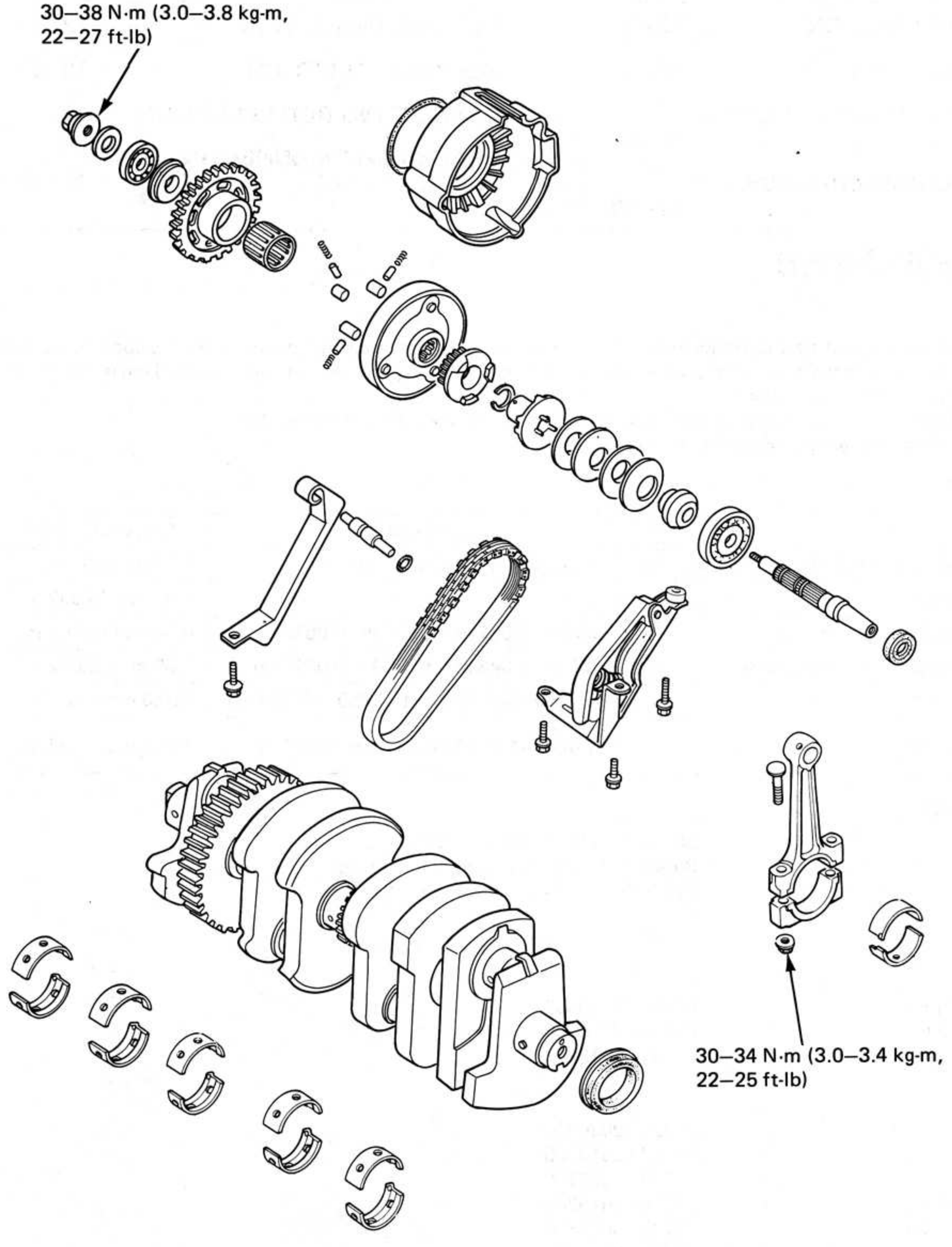

30–38 N·m (3.0–3.8 kg-m,
22–27 ft-lb)
30–34 N·m (3.0–3.4 kg-m,
22–25 ft-lb)

# SERVICE INFORMATION

**GENERAL**

- All bearing inserts are a select fit and are identified by colour codes. Select replacement bearings from the code tables. After installing new bearings, recheck them with plastigauge to verify clearance. Apply molybdenum disulfied grease the main journals and crankpins during assembly.
- The crankcase assembly must be separated (section 10) to service the crankshaft and starter clutch.
- Refer to section 18 for starter system troubleshooting.

**SPECIFICATONS**

| | | STANDARD | SERVICE LIMIT |
|---|---|---|---|
| Crankshaft | Connecting rod big end side clearance | 0.05-0.20 mm (0.002-0.008 in) | 0.3 mm (0.01 in) |
| | Runout | | 0.05 mm (0.002 in) |
| | Crankpin oil clearance | 0.024-0.057 mm (0.0009-0.0022 in) | 0.06 mm (0.002 in) |
| | Main journal oil clearance | 0.019-0.043 mm (0.0007-0.0017 in) | 0.05 mm (0.002 in) |
| Cam chain | Length | 336.55-337.00 mm (13.250-13.268 in) | 320.0 mm (12.60 in) |
| Alternator chain | Length | 149.00-149.20 mm (5.866-5.874 in) | 150.5 mm (5.93 in) |

**TORQUE VALUES**

| | |
|---|---|
| Alternator shaft | 30-38 N.m (3.0-3.8 kg.m, 22-27 ft.lb) |
| Connecting rod cap | 30-34 N.m (3.0-3.4 kg.m, 22-25 ft.lb) |
| Main bearing | 21-25 N.m (2.1-2.5 kg.m, 15-18 ft.lb) |

**TOOLS**

**Special**

| | |
|---|---|
| Bearing remover, 17 mm | 07936-3710300 |
| Bearing remover handle | 07936-3710100 |
| Bearing remover weight | 07741-0010201 |

**Common**

| | |
|---|---|
| Universal holder | 07725-0030000 |
| Driver | 07749-0010000 |
| Driver | 07746-0020100 |
| Attachment, 37 x 40 | 07746-0010200 |
| Attachment, 42 x 47 | 07746-0010300 |
| Pilot, 17 mm | 07746-0040400 |
| Pilot, 20 mm | 07746-0040500 |
| Attachment, I.D. 20 mm | 07746-0020400 |

# TROUBLESHOOTING

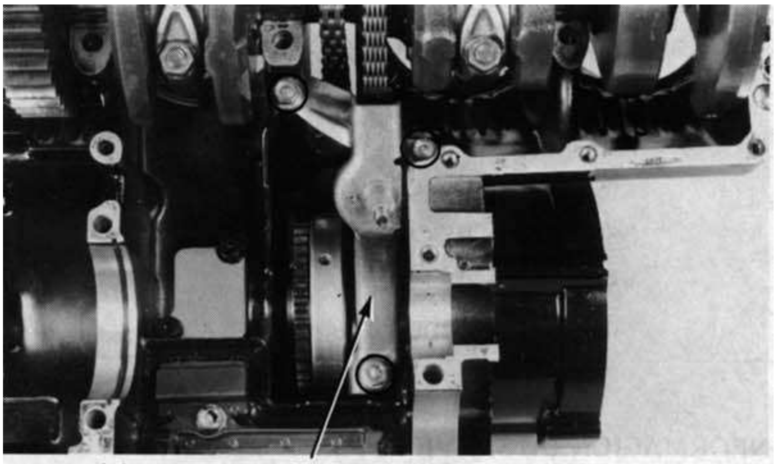
**Excessive noise**

1 Worn main journal bearing
2 Worn crank pin bearing

# STARTER CLUTCH / ALTERNATOR SHAFT

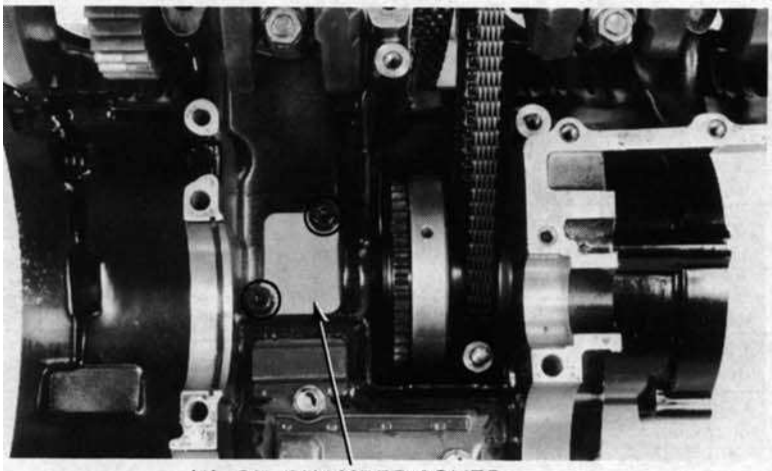
**STARTER CLUTCH/ALTERNATOR SHAFT REMOVAL**

Separate the crankcase (page 10-2)
Remove the transmission.
Remove the alternator drive chain tensioner by removing the two bolts and nut.
Remove the spacer collar from the tensioner mounting stud.



(1) ALTERNATOR DRIVE CHAIN TENSIONER

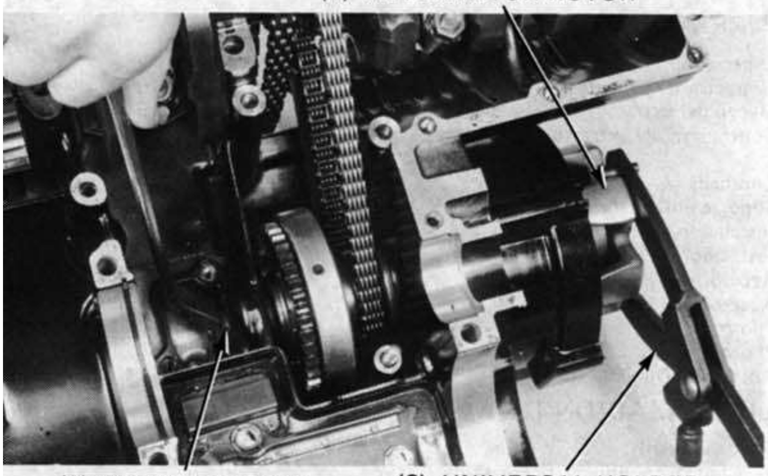
Remove the two socket bolts and the oil chamber cover.



(1) OIL CHAMBER COVER

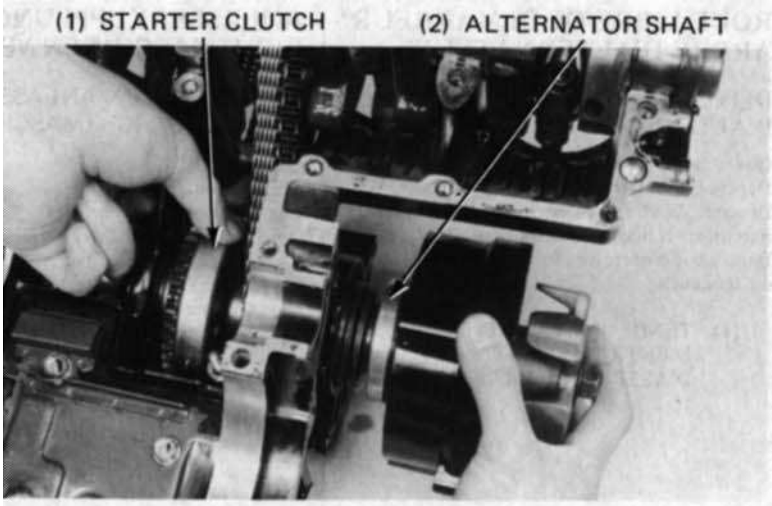
Temporarily install the alternator rotors and hold the rotor with universal holder.
Remove the alternator shaft mounting nut.

(1) ALTERNATOR ROTOR



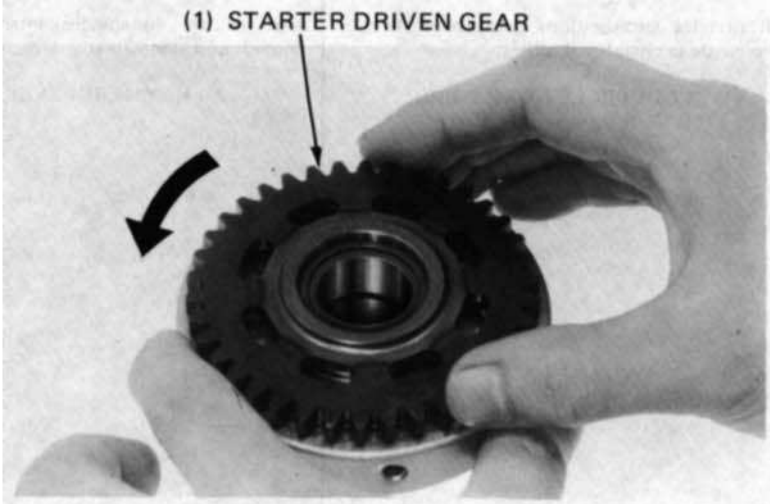
(2) MOUNTING NUT (3) UNIVERSAL HOLDER 07725–0030000

Pull the alternator shaft out of the upper crankcase and remove the starter clutch, spacer and alternator driven sprocket.

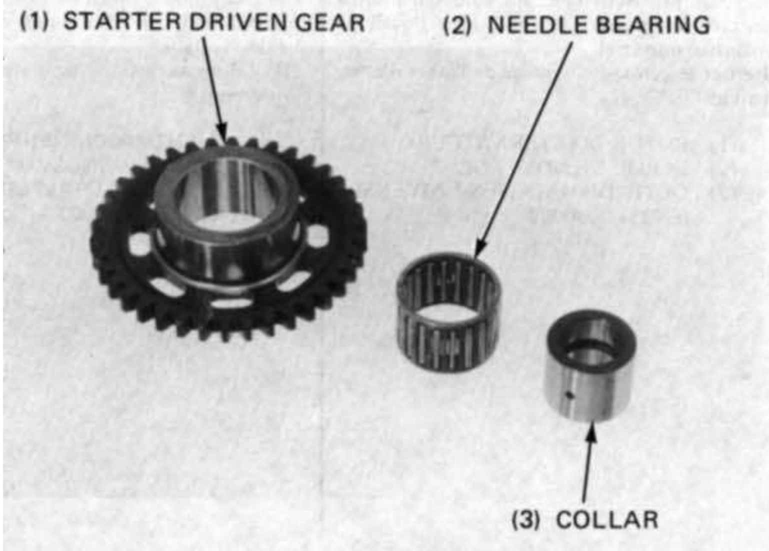



**STARTER CLUTCH INSPECTION**

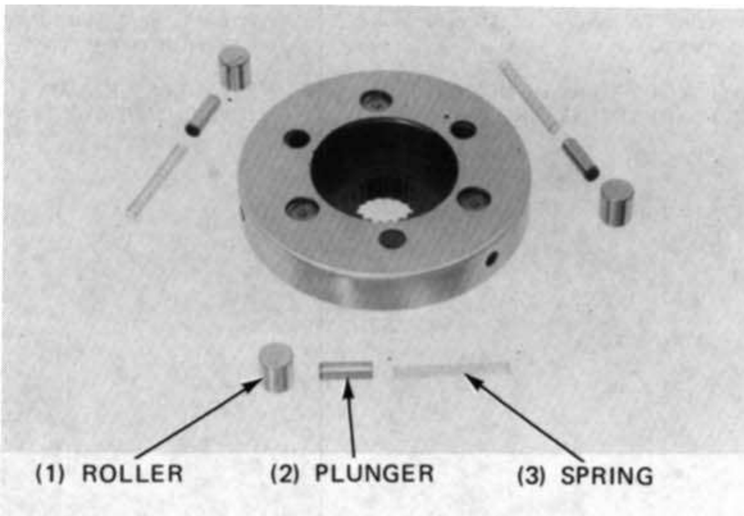
Check the starter clutch for smooth operation by turning the starter driven gear. The starter driven gear should turn counterclockwise freely and should not turn clockwise.




Remove the starter driven gear, needle bearing and collar from the starter clutch. Check the driven gear, needle bearing and collar for wear or damage.



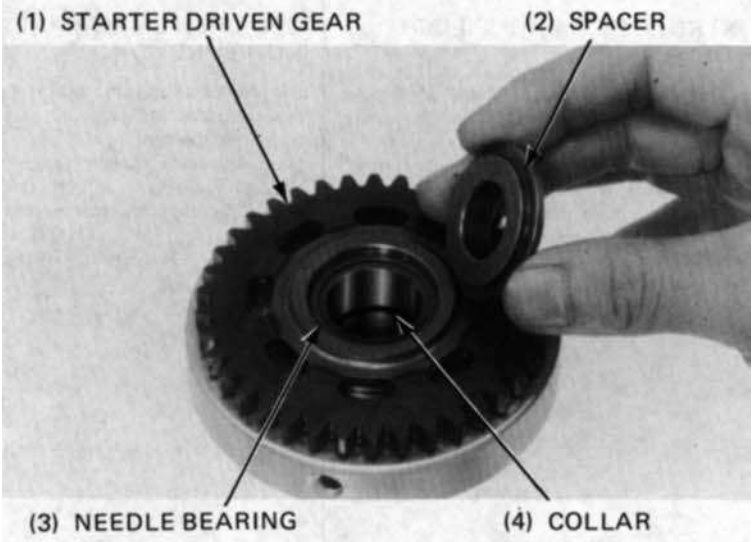
Remove the rollers, plungers and springs from the starter clutch and check them for wear or damage.




**STARTER CLUTCH ASEMBLY**

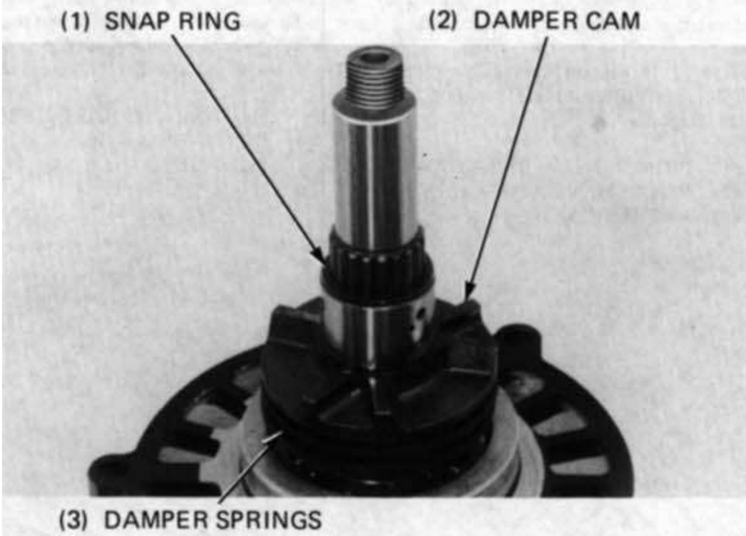
Install the springs, plungers and rollers into the starter clutch.

Install the needle bearing and collar into the starter driven gear and install them into the starter clutch.
Install the spacer onto the starter driven gear.




**ALTERNATOR SHAFT DISASSEMBLY**

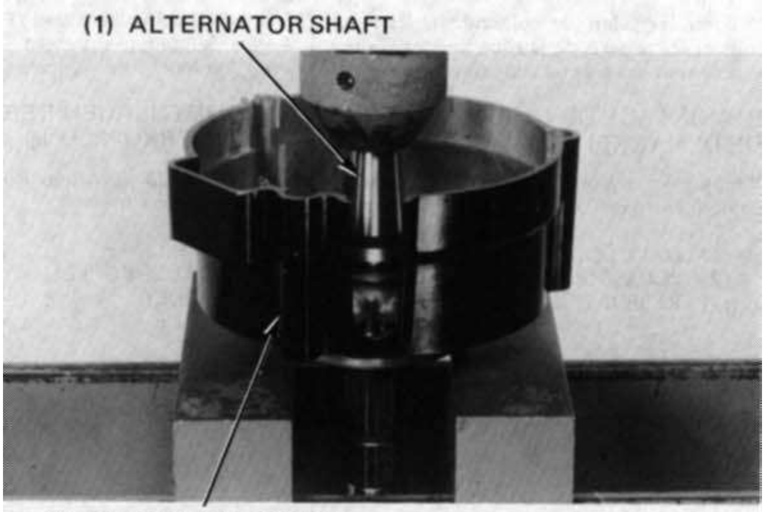
Remove the snap ring, damper cam and damper springs from the alternator shaft.



Place the alternator shaft/case in the hydraulic press with the case supported.

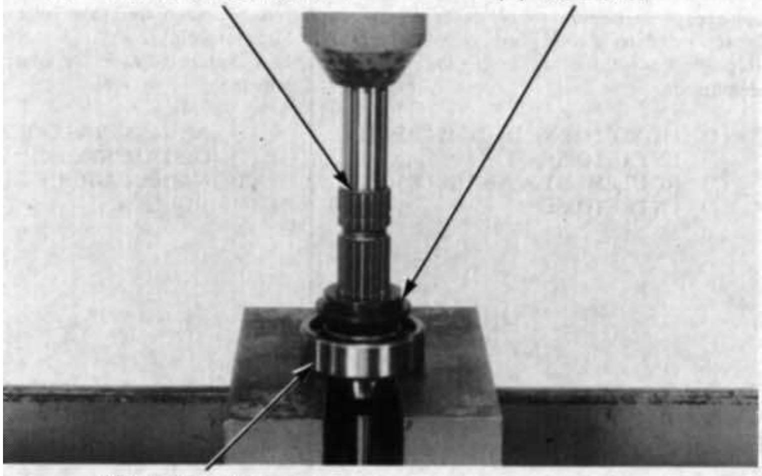
**NOTE:**
**Do not support the bearing as the bearing is pressed out together with the shaft.**

Press the alternator shaft out of the alternator case.

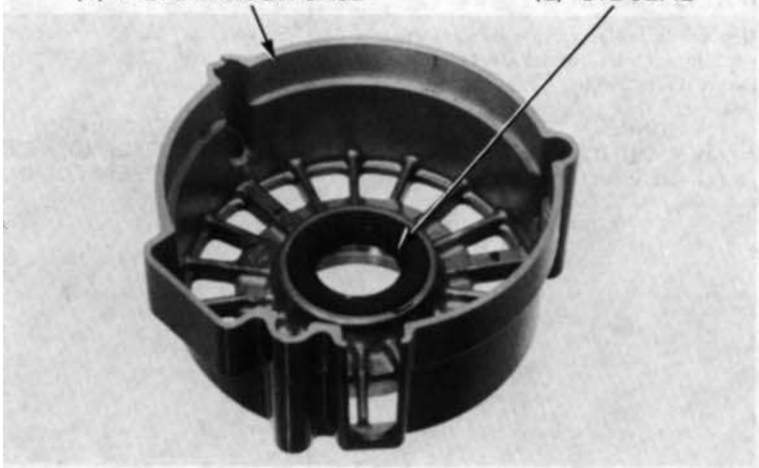



Place the alternator shaft in the press and press the shaft out of the bearing and collar.

**NOTE:**
**Never reinstall old bearing; once the bearing is removed, it must be replaced with a new one.**

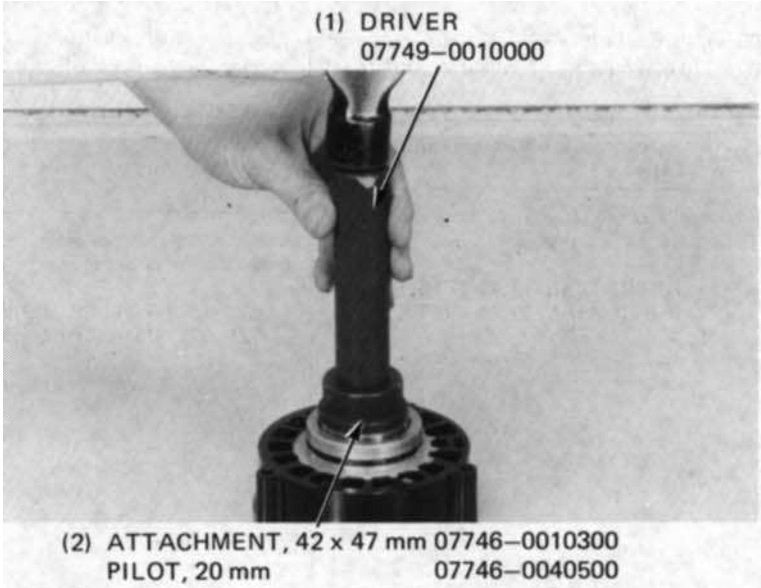



Remove the oil seal from the alternator case.

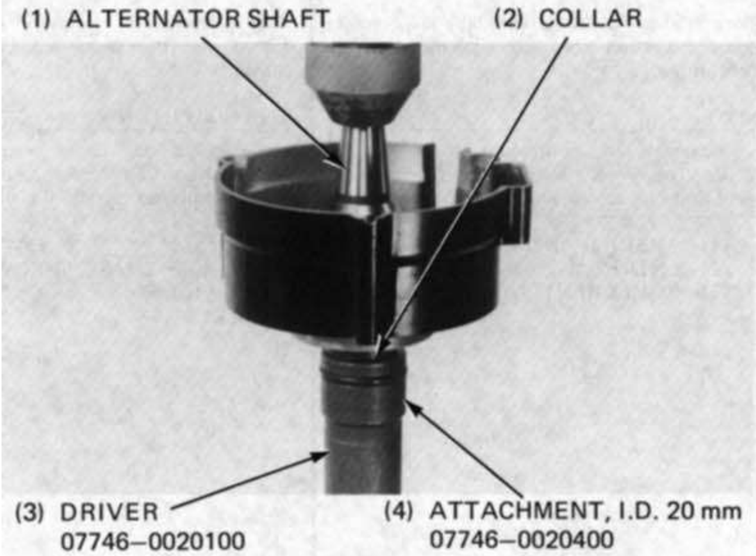


## ALTERNATOR SHAFT ASSEMBLY

Drive a new bearing in the alternator case.
Install a new oil seal into the case.

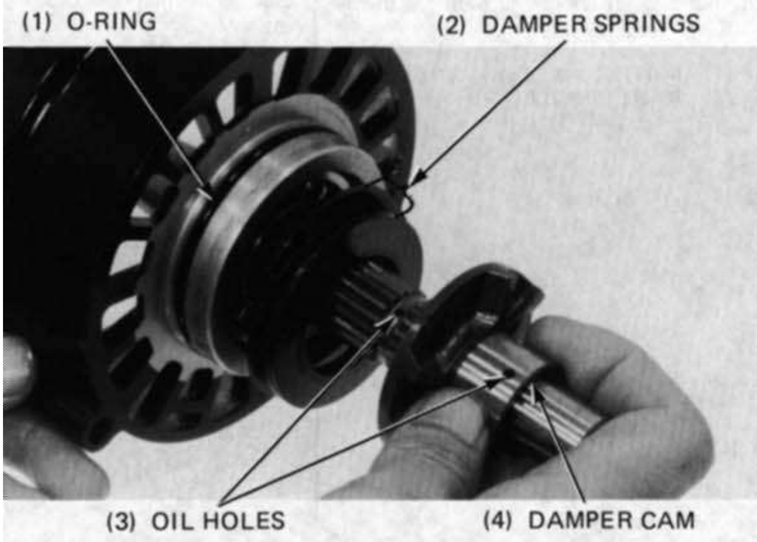



Support the collar and case bearing with special tools and press the alternator shaft into them.



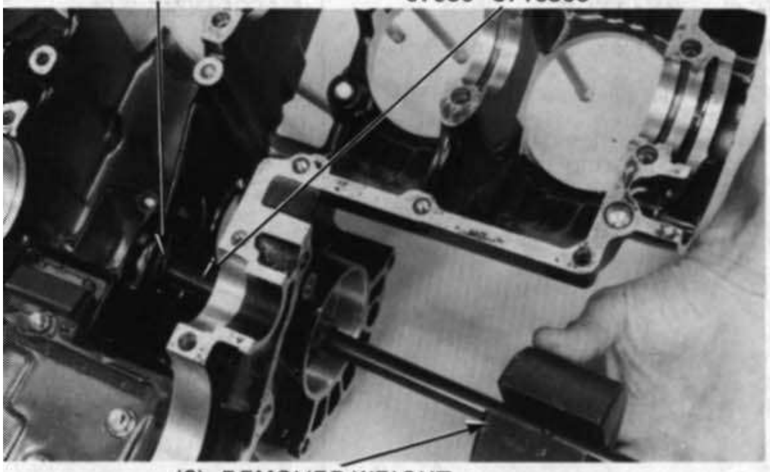

Install the four damper springs with the dished faces facing each other as shown.
Install the damper cam onto the alternator shaft, aligning the oil holes in the damper cam and shaft.

Secure the damper cam with the snap ring.
Install a new O-ring into the groove in the alternator case.

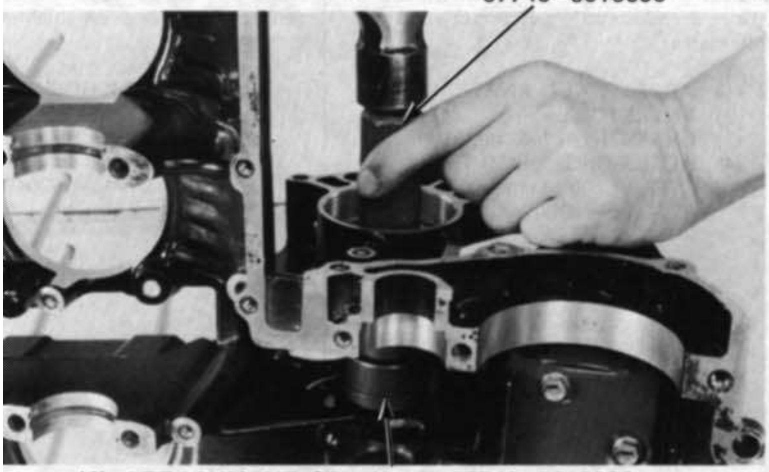


**ALTERNATOR SHAFT BEARING REPLACEMENT (CRANKCASE SIDE)**

Remove the crankshaft (page 12-10)
Remove the alternator shaft bearing with special tools.

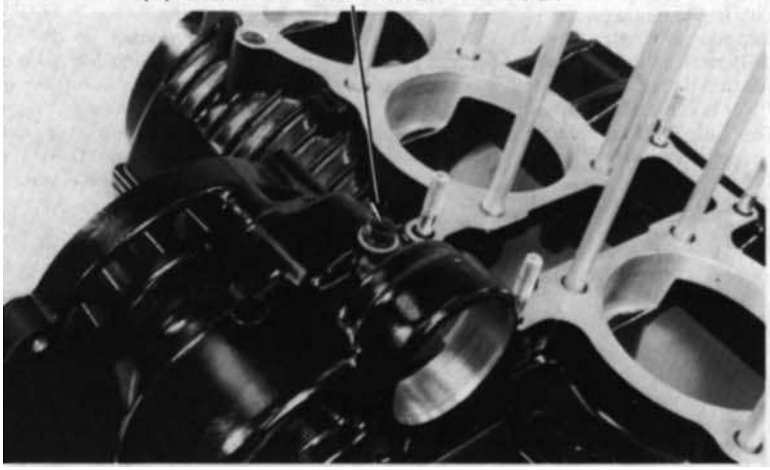



Drive a new bearing in the crankcase.
Install the crankshaft (page 12-17)

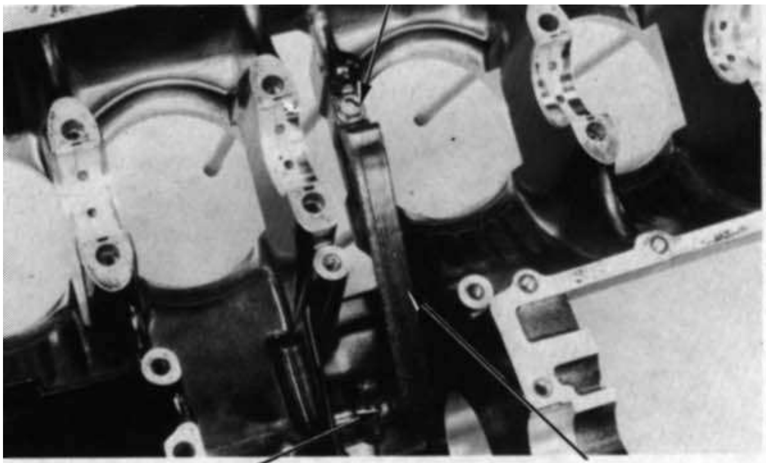



**ALTERNATOR DRIVE CHAIN SLIPPER REPLACEMENT**

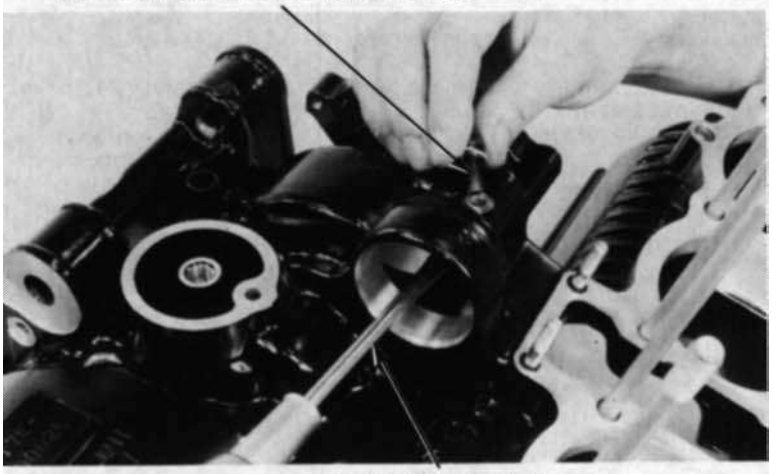
Remove the crankshaft (12-10)
Remove the slipper pin attaching bolt.



Remove the slipper attaching bolt, pull the slipper pin out and remove the slipper.
Replace the slipper if it is damaged.
Install the slipper on the upper crankcase and install the slipper pin.
Apply thread lock agent to the attaching bolt threads and install it.

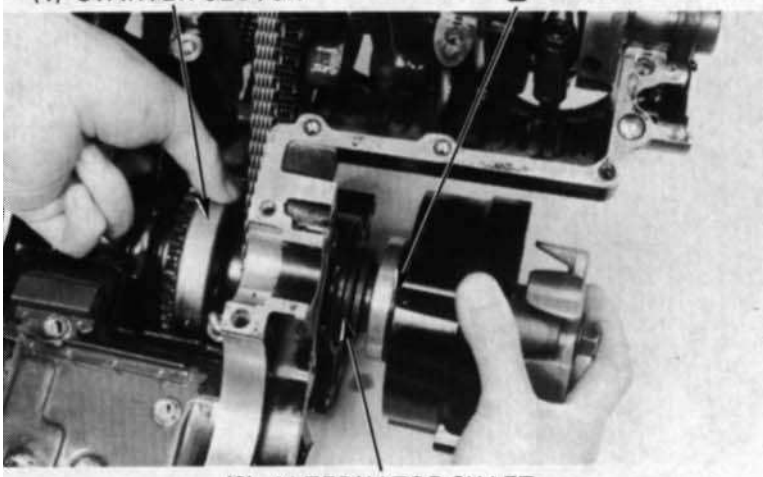



Align the bolt holes in the upper crankcase and slipper pin by turning the pin with a screwdriver.
Apply thread lock agent to the threads of the slipper pin attaching bolt and install it.
Install the crankshaft (page 12-17)



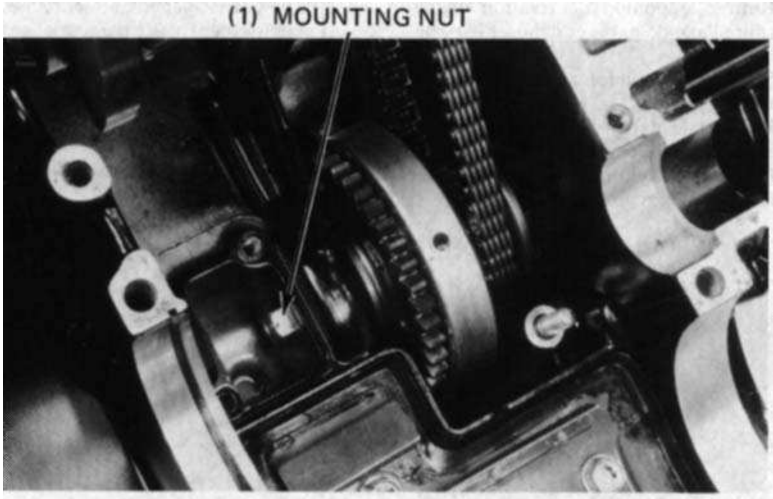

**STARTER CLUTCH - ALTERNATOR SHAFT INSTALLATION**

Place the alternator drive chain over the alternator driven sprocket.
Coat the alternator case O-ring with clean engine oil and insert the alternator shaft into the upper crankcase through the alternator driven sprocket and starter clutch.

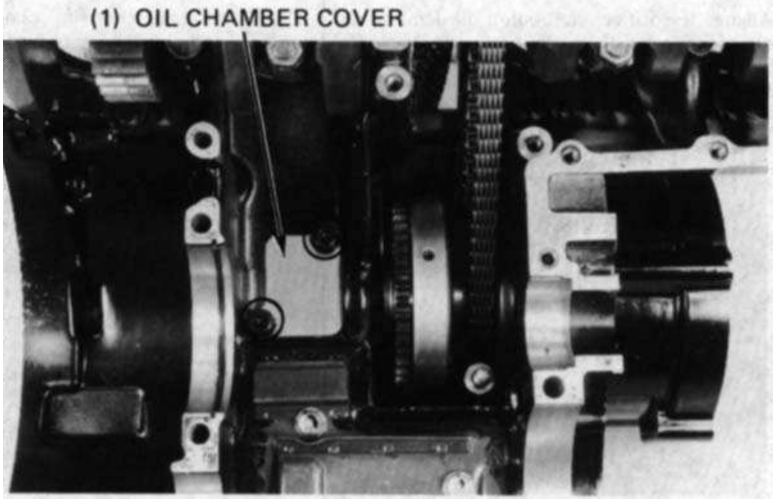


Temporarily install the alternator rotors and hold the rotor with universal holder (07725-0030000). Install and tighten the alternator shaft mounting nut.

**TORQUE: 30-38 Nm (3.0-3.8 kg.m, 22-27 ft.lb)**

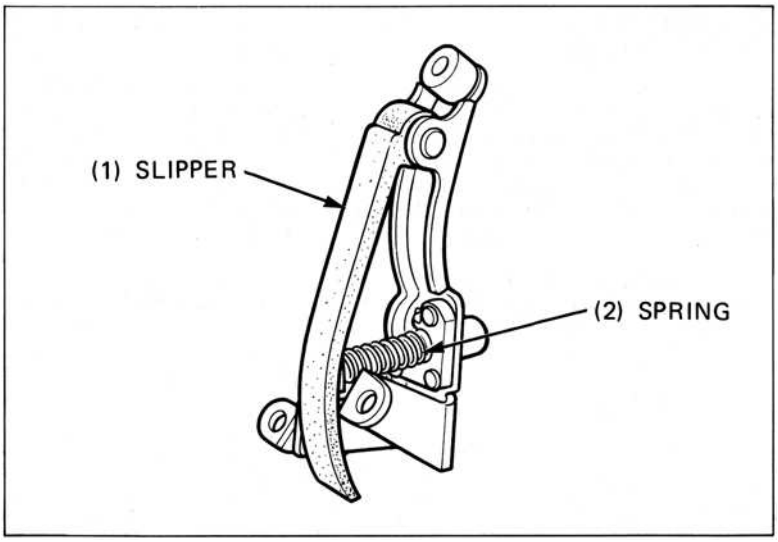



Apply thread lock agent to the socket bolt threads and install the oil chamber cover with the socket bolts.

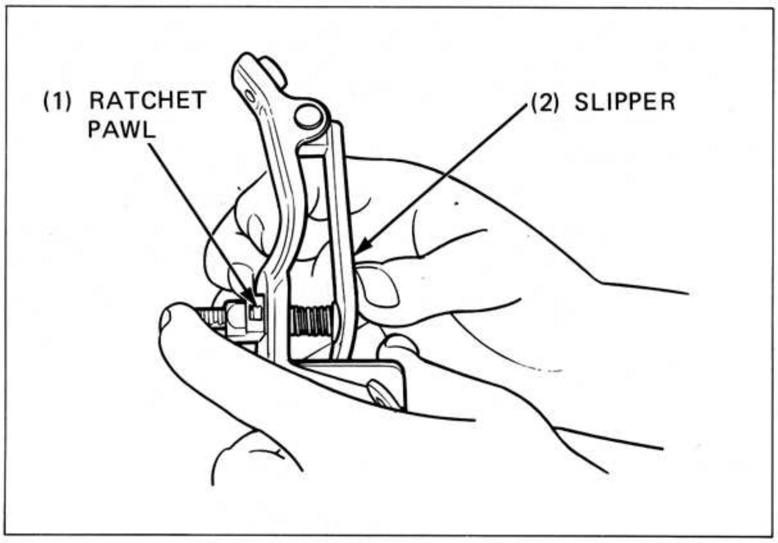



Check the alternator drive chain tensioner spring and slipper for wear or damage.

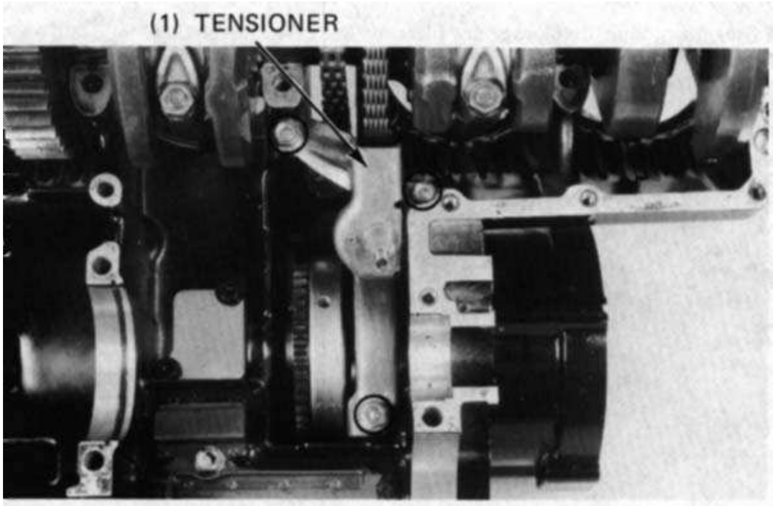
Replace the tensioner if necessary.



Release the ratchet paw] by pushing it and push the tensioner slipper on for minimum tension and hold it until the tensioner is installed in position.



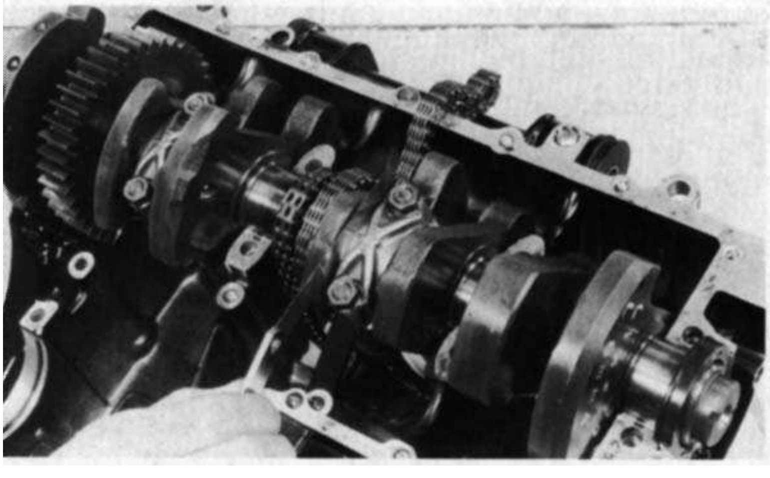

Apply thread lock agent to the threads of the attaching bolts and nut, and install the tensioner over the drive chain and on the crankcase with the bolts and nut.



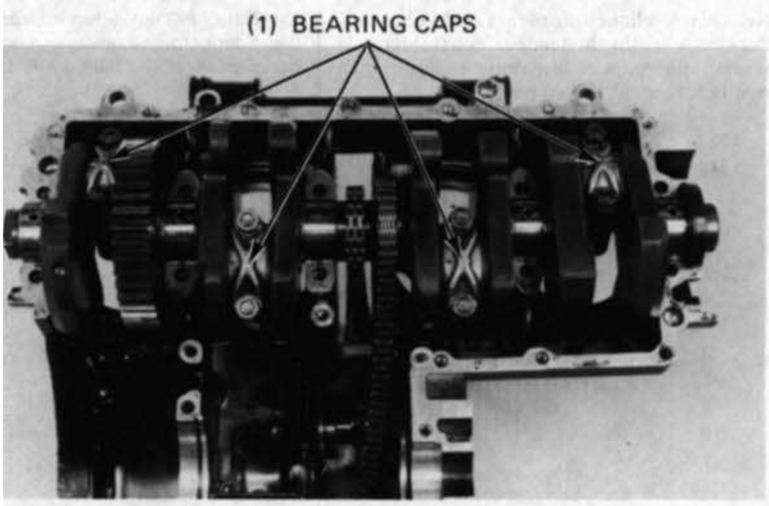

## CONNECTING ROD/ CRANKSHAFT REMOVAL

Separate the crankcase (page 10-2).
Remove the alternator shaft (page 12-2).
Check the connecting rod side clearance with a feeler gauge.

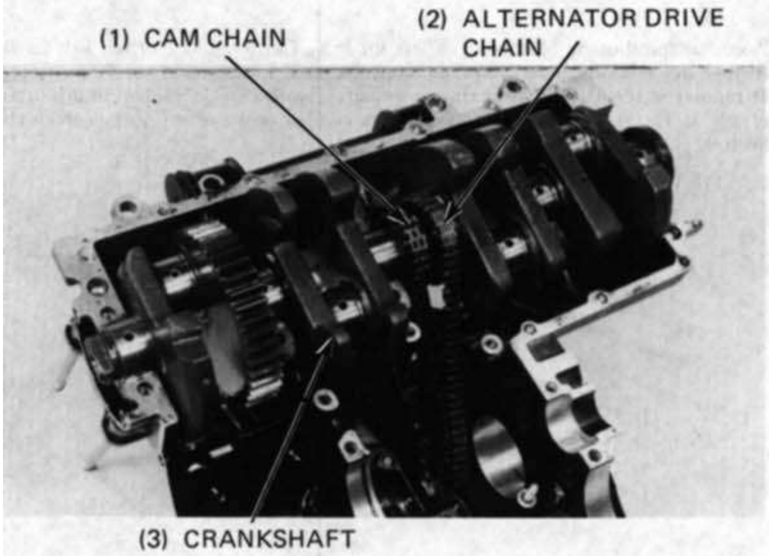
**SERVICE LIMIT: 0.3 mm (0.01 in)**



Remove the bearing cap nuts, bearing caps and connecting rods.
Mark the rods, bearings and bearing caps to indicate their cylinder position for correct reassembly.




Remove the crankshaft, cam chain and alternator drive chain.

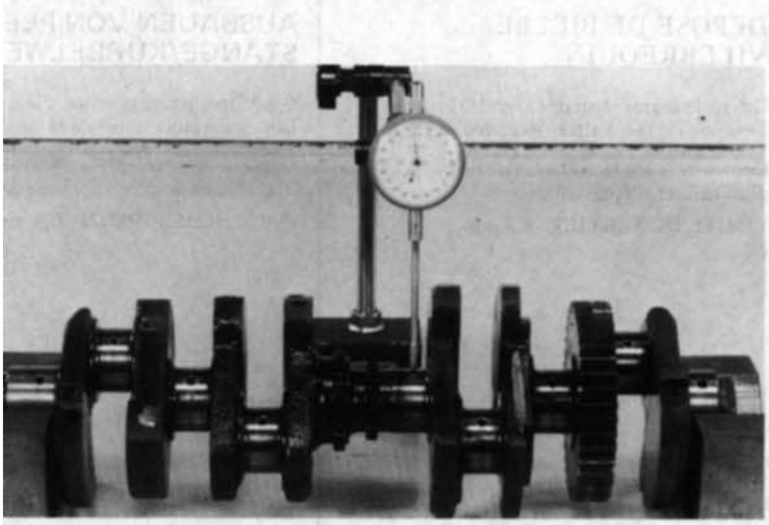



**INSPECTION**

**CRANKSHAFT RUNOUT**

Remove the cam and alternator chains.
Place the crankshaft on a stand or V-blocks

Set a dial indicator on the centre main journal of the crankshaft.
Rotate the crankshaft two revolutions and read runout at the centre journal.

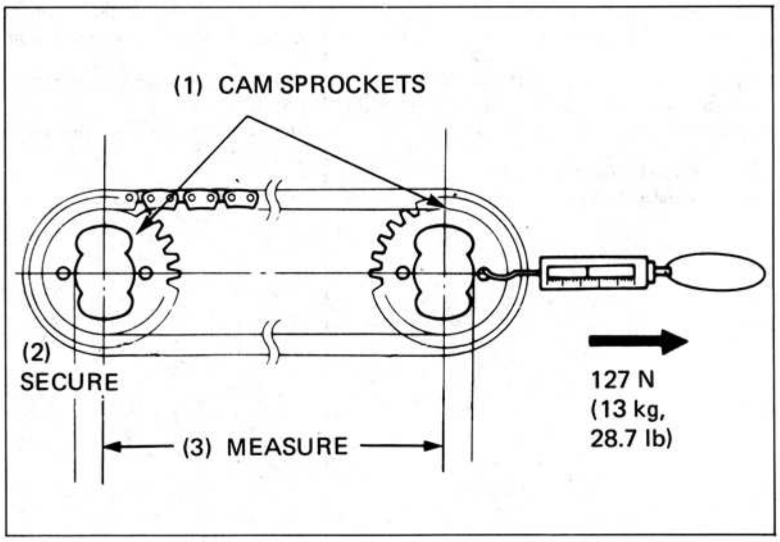


**SERVICE LIMIT: 0.05 mm (0.002 in)**

### CAM CHAIN LENGTH MEASUREMENT

Place the cam chain over the intake and exhaust cam shaft sprockets with the bolt holes positioned as shown. Secure one sprocket.
Apply 127 N (13 kg, 29 lb) of tension with a spring scale to the other sprocket.
Measure the chain length between the sprocket centres.

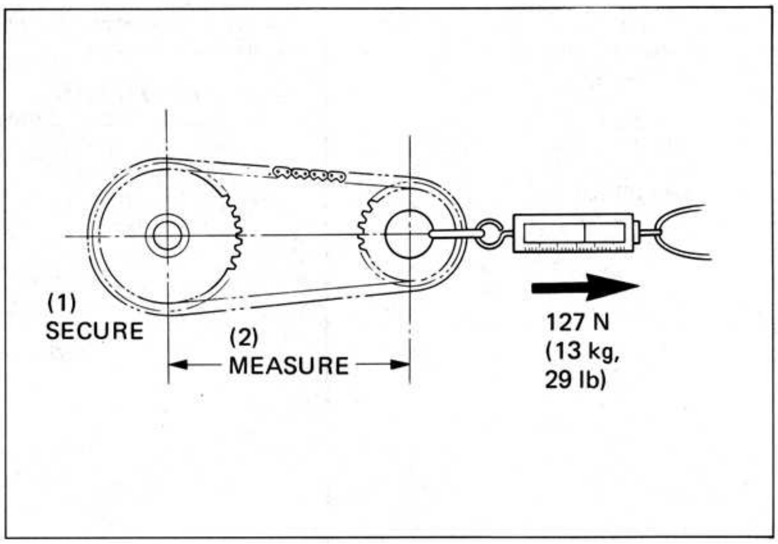
**SERVICE LIMIT: 320.0 mm (12.60 in)**




### ALTERNATOR DRIVE CHAIN LENGTH

Place the alternator drive chain over the alternator drive and driven sprockets.
Secure the crankshaft.
Apply 127 M (13 kg, 29 lbs) of tension with a spring scale to the driven sprocket.
Measure the chain length between the sprocket centres.

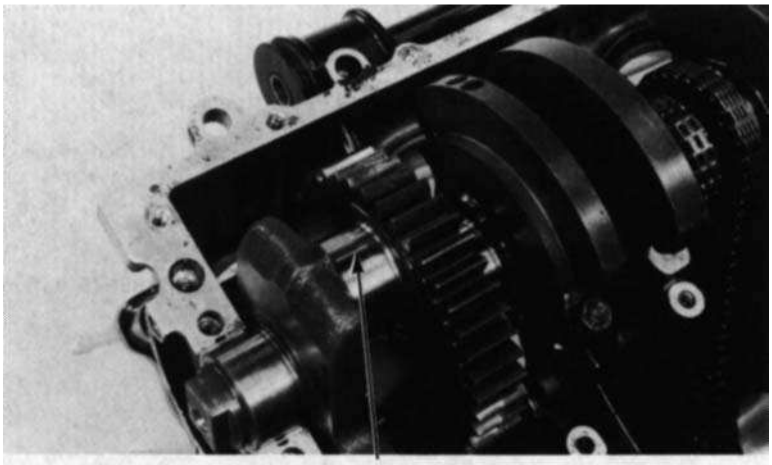
**SERVICE LIMIT: 150.5 mm (5.93 in)**




## BEARING INSPECTION

### CONNECTING RODS

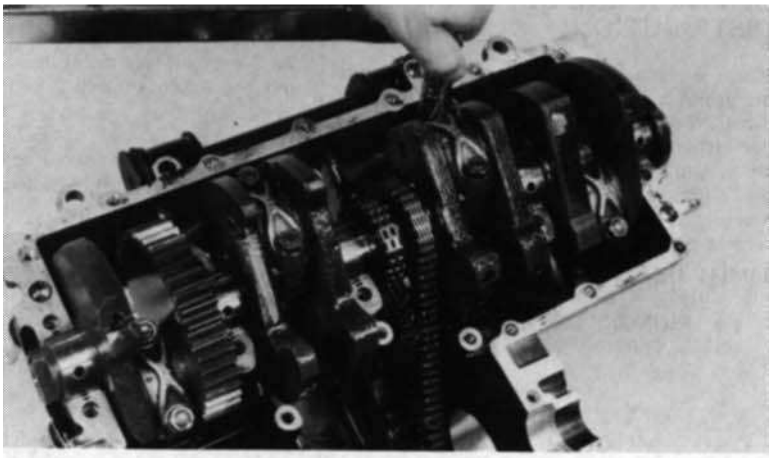
Inspect the bearing inserts for damage or separation.
Clean all oil from the bearing inserts and crankpins.
Put a piece of plastigauge on each crankpin avoiding the oil hole.


(1) PLASTIGAUGE

Install the bearing caps and rods on the correct crankpins, and tighten them evenly.

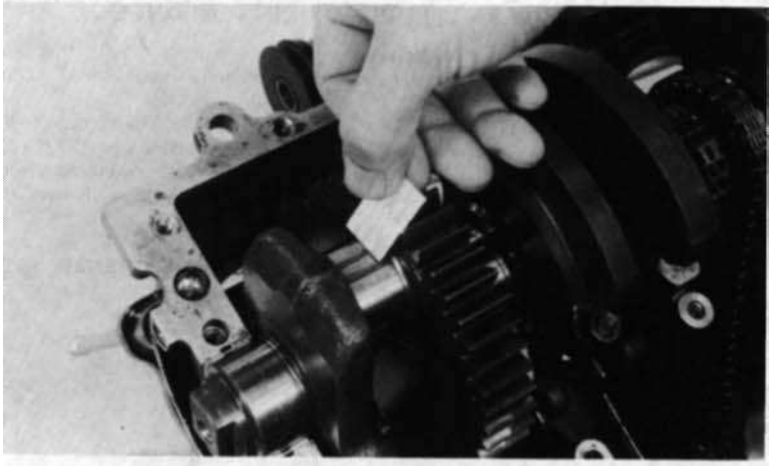
**TORQUE: 30-34 Nm**
**(3.0-3.4 kg.m, 22-25 ft.lb)**

**NOTE:**
**Do not rotate the crankshaft during inspection.**



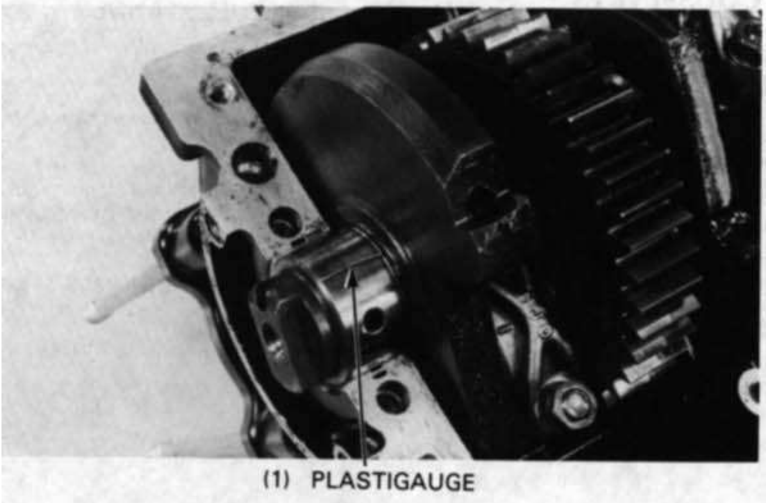
Remove the caps and measure the compressed plastigauge on each crankpin.

**OIL CLEARANCE SERVICE LIMIT:**
**0.06 mm (0.002 in)**



**MAIN BEARINGS**

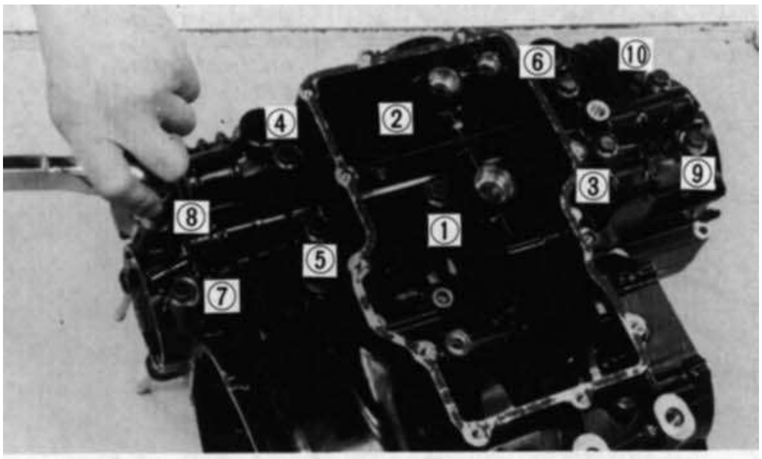
Inspect the bearing inserts for damage or separation.
Clean all oil from the bearing inserts and journals.
Put a piece of plastigauge on each journal, avoiding the oil holes.



Install the main bearings on the correct journals on the lower crankcase and tighten them evenly in the sequence shown and in 2-3 steps.

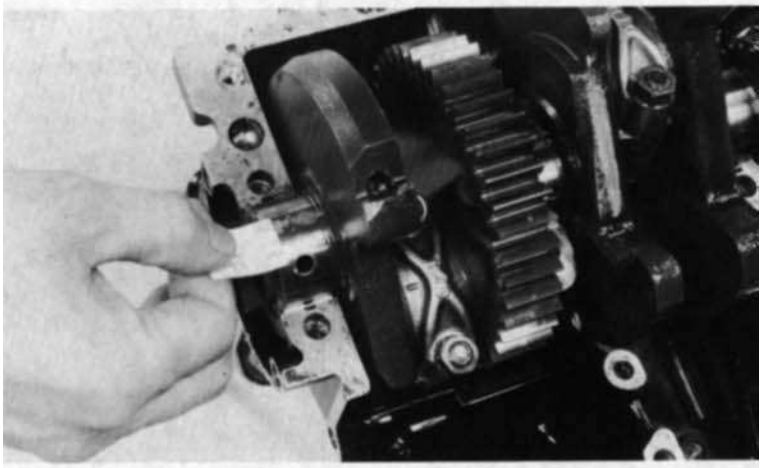
**TORQUE: 21-25 Nm**
**(2.1-2.5 kg.m, 15-18 ft.lb)**

**NOTE:**
**Do not rotate the crankshaft during inspection.**




Remove the lower crankcase and measure the compressed plastigauge on each journal.

**OIL CLEARANCE SERVICE LIMIT:**
**0.05 mm (0.002 in)**

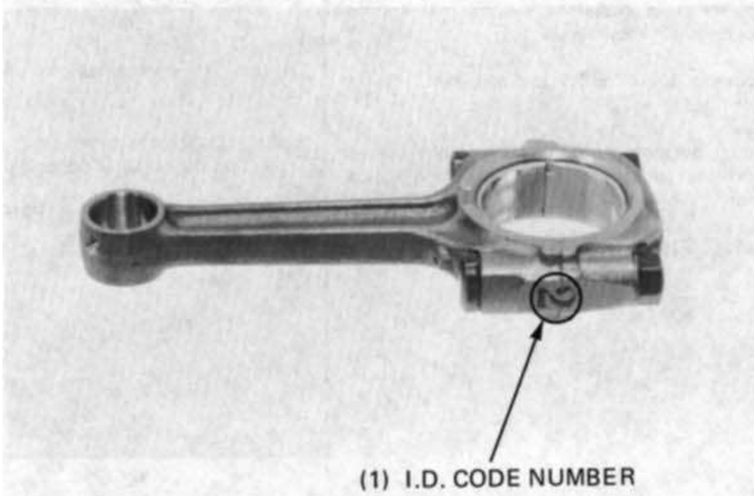


## BEARING SELECTION

If rod bearing clearance is beyond tolerance, select replacement bearings as follows:

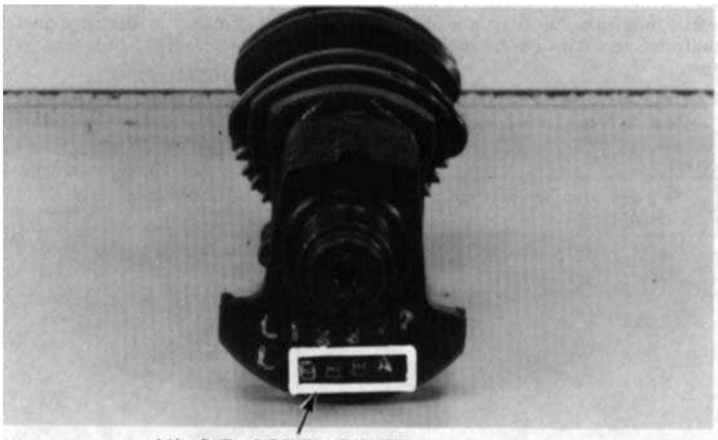
**CONNECTING ROD BEARING INSERTS**

Determine and record the corresponding rod I.D. code number.



Determine and record the corresponding crankpin O.D. code number (or measure the crankpin O.D.).

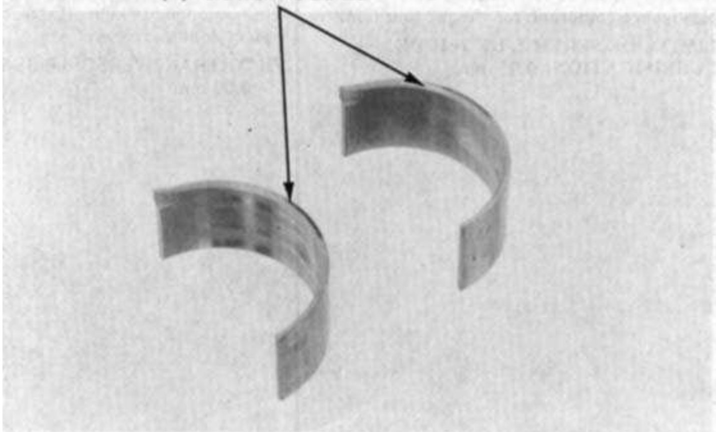
**NOTE:**
**The letter A or B on the outside crankshaft weight is the code for each crankpin O.D. from left-to-right.**



(1) O.D. CODE LETTER

Cross reference the crankpin and connecting rod codes to determine the replacement bearing colour

(1) COLOR CODES

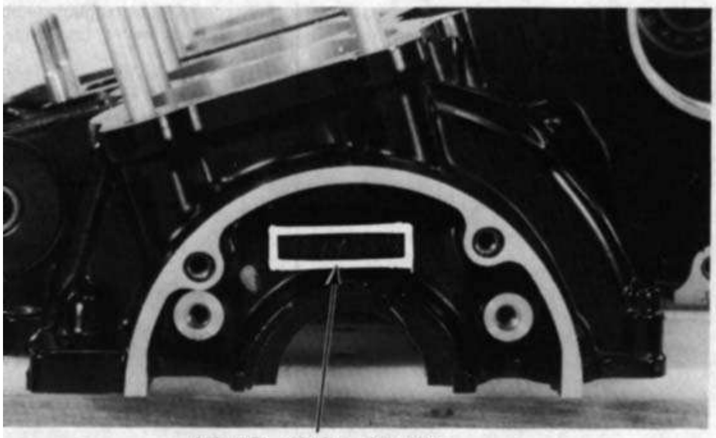


<table>
<tr><td colspan="3" rowspan="3"></td><td colspan="2">CRANKPIN O.D. CODE LETTER</td></tr>
<tr><td>A</td><td>B</td></tr>
<tr><td>35.992-36.000 mm (1.4170-1.4173 in)</td><td>35.984-35.992 mm (1.4167-1.4170 in)</td></tr>
<tr><td rowspan="2">CONNECTING ROD I.D. CODE NUMBER</td><td>1</td><td>39.000-39.008 mm (1.5354-1.5357 in)</td><td>Yellow</td><td>Green</td></tr>
<tr><td>2</td><td>39.008-39.016 mm (1.5357-1.5361 in)</td><td>Green</td><td>Brown</td></tr>
</table>

**BEARING INSERT THICKNESS:**
**Brown: 1.494-1.498 mm (0.0588-0.0590 in)**
**Green: 1.490-1.494 mm (0.0587-0.0588 in)**
**Yellow: 1.486-1.490 mm (0.0585-0.0587 in)**

**MAIN BEARING**

Determine and record crankcase I.D. code number on the upper crankcase.



(1) I.D. CODE LETTER

**NOTE:**
**The letters A or B on the upper crankcase are the codes for the main journal I.D.'s from left-to-right.**

Determine and record the corresponding main journal I.D. code letters (or measure the main journal O.D.).

**NOTE:**
**The letters 1 or 2 on the crank weight is the code for the main journal O.D.'s from left-to-right.**

Cross reference the case and journal codes to determine the replacement bearings.

| | | | MAIN JOURNAL O.D. CODE NUMBER | |
|---|---|---|---|---|
| | | | A | B |
| | | | 35.992-36.000 mm (1.4170-1.4173 in) | 35.984-35.992 mm (1.4167-1.4170 in) |
| CRANKCASE I.D. CODE LETTER | 1 | 39.000-39.008 mm (1.5354-1.5357 in) | Pink | Yellow |
| | 2 | 39.008-39.016 mm (1.5357-1.5361 in) | Yellow | Green |

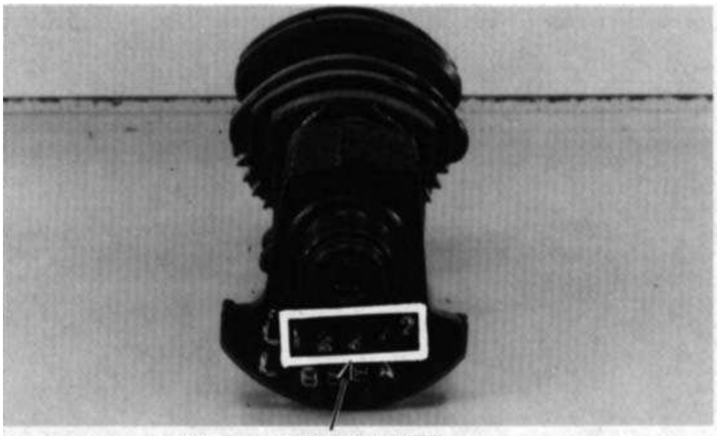
**BEARING INSERT THICKNESS:**
**Green:1.504-1.508 mm (0.0592-0.0594 in)**
**Yellow:1.500-1.504 mm (0.0591-0.0592 in)**
**Pink: 1.496-1.500 mm (0.0589-0.0591 in)**

## CONNECTING ROD SELECTION

When replacing the connecting rods, select the rods accordance with the weight code as shown below.

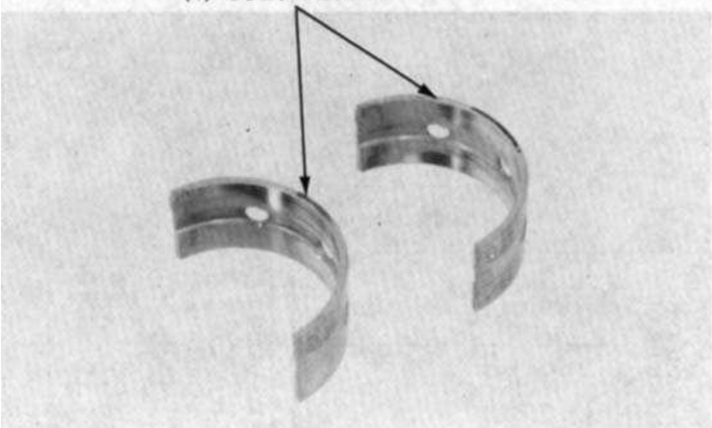
| Factory set code | Available code |
|---|---|
| A | B |
| B | B |
| C | D |
| D | D |
| E | D |

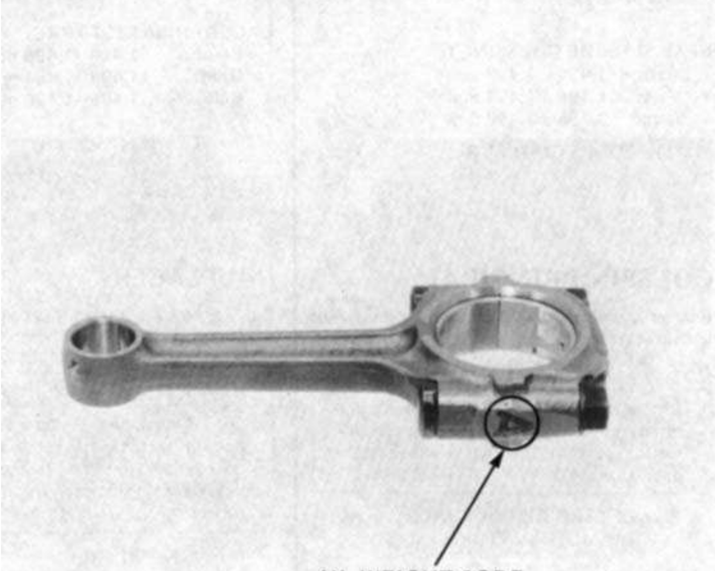
**NOTE:**
**Do not select the connecting rod with the different weight code of 2 rank.**



(1) O.D. CODE NUMBER

(1) COLOR CODES

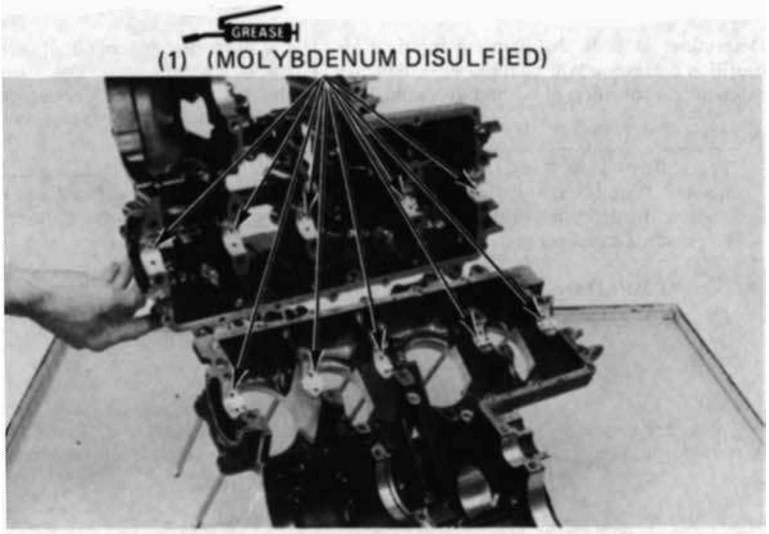




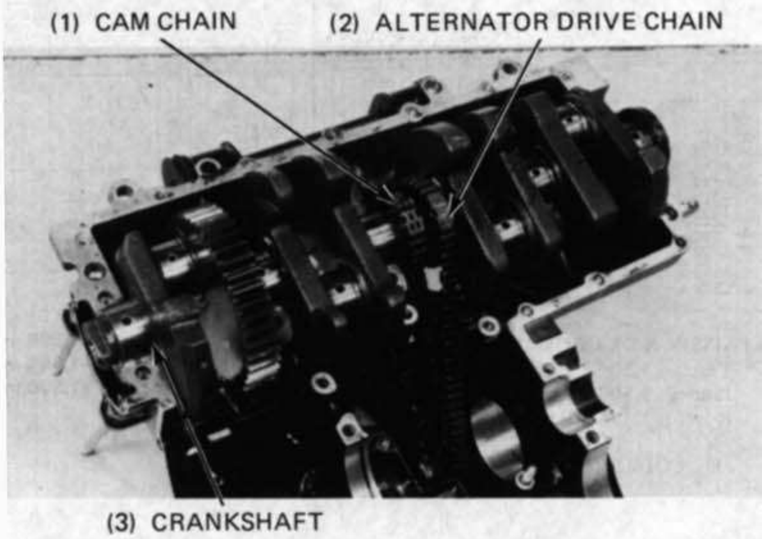
(1) WEIGHT CODE

## CRANKSHAFT/CONNECTING ROD INSTALLATION

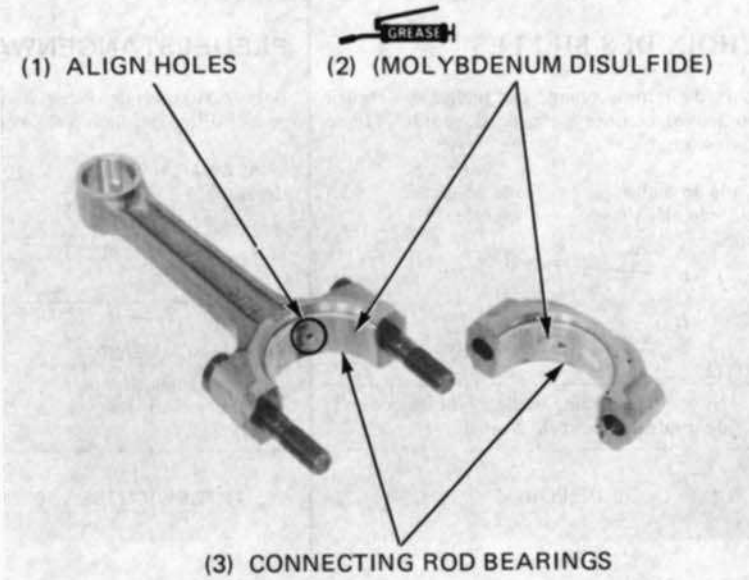
Install the main bearings into the upper and lower crankcases.
Apply molybdenum disulfide grease to the upper and lower main bearings.



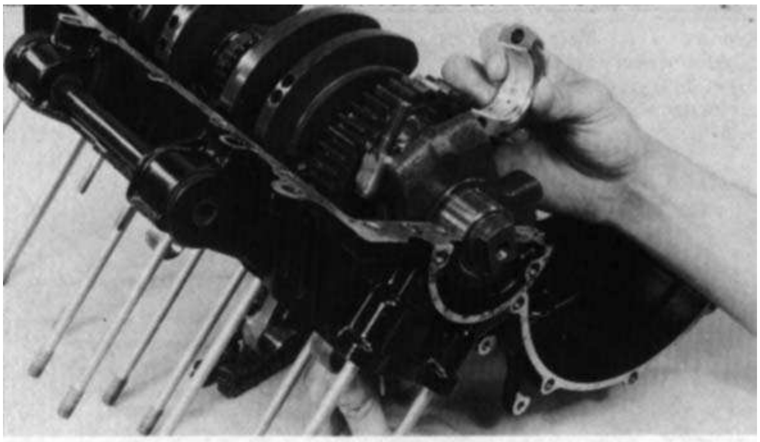

Install the cam chain and alternator drive chain over the crankshaft. Install the crankshaft onto the upper crankcase.




Align the hole in the connecting rod bearing insert with the hole in the connecting rod and install the insert. Install the connecting rod cap bearing insert.
Apply molybdenum disulfide grease to the connecting rod bearings.

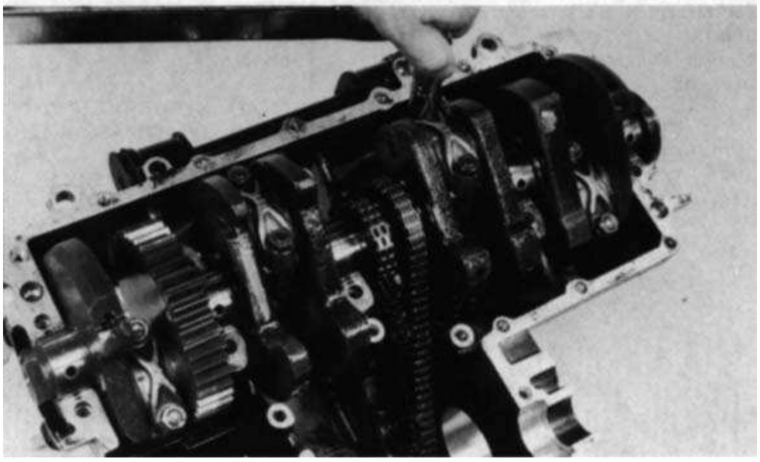


Apply molybdenum disulfide grease to the threads and flanges of the connecting rod cap nuts, and install them.



**NOTE:**
- **Be sure the connecting rods are installed in their correct positions and the oil holes point to the rear (intake side).**
- **Cross reference the connecting rod and cap I.D. codes to insure correct assembly.**

Tighten the connecting rod cap nuts in 2-3 steps.



**TORQUE. 30-34 Nm**
**(3.0-3.4 kg.m, 22-25 ft.lb)**

**NOTE:**
**After tightening the nuts, check that the connecting rod moves freely without binding.**

Install the alternator shaft (page 12-8).
Assemble the crankcase (section 10).